

Impact Of Race On Baseline Asthma Severity and Response To Biologic Therapy In The First 12 Months Post Biologic Initiation

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HIGHLIGHTS

- At biologic initiation, **Black or African American/Other** patients had **worse lung function** than White patients.
- However, the **difference narrowed between these two groups** significantly over the first year of treatment.
- Interventions to **reduce treatment gaps and promote pharmacoequity** in patients with moderate-to-severe asthma are needed.

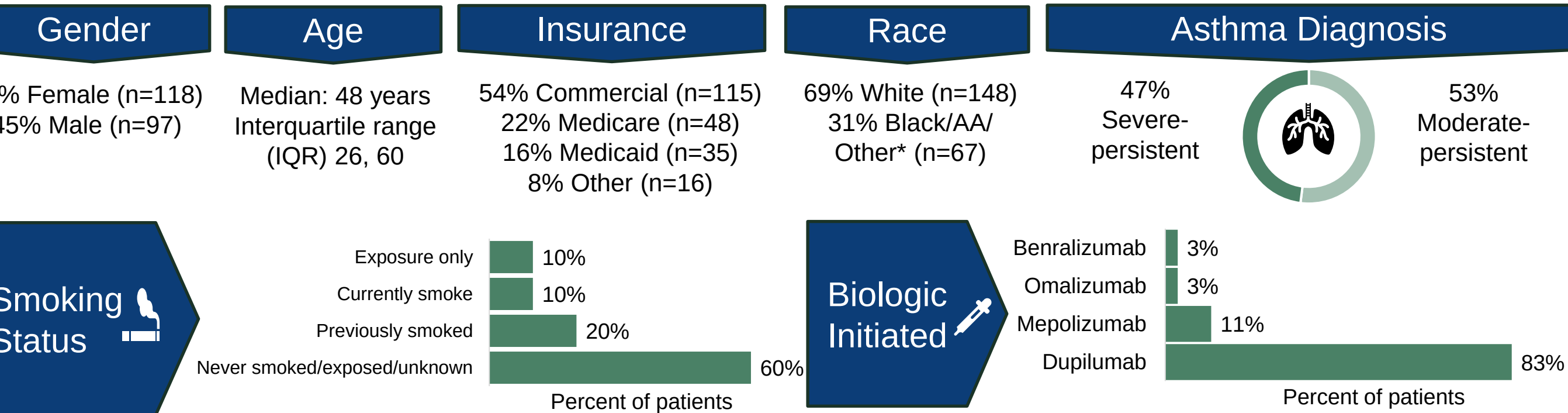
PURPOSE

The purpose of the study was to identify racial and socioeconomic factors that may impact outcomes in patients with moderate-to-severe asthma before and within 12 months of biologic initiation.

METHODS

Design	- Retrospective cohort study Enrollment: 1/1/2019-6/30/2022 with a12-month follow up through 6/30/2023
Sample	Inclusion criteria: Patients prescribed a new biologic (benralizumab, dupilumab, mepolizumab, and omalizumab) between 1/1/2019 and 6/30/2022 Exclusion criteria: Indication other than moderate to severe asthma; Biologic not prescribed by a Vanderbilt Allergy/ Immunology/ Pulmonary Clinic provider
Analysis	Primary Outcome: Race-neutral ppFEV₁ (forced expiratory volume in one second, percent predicted) calculated using pre-bronchodilator FEV₁ measured approximately within the first year post biologic initiation as a measure of efficacy/response to biologic therapy in the first year of follow-up. Secondary Outcomes: Biologic medication adherence and gap days. Adherence was calculated using proportion of days covered (PDC). Gap days were defined as time from medication exhaustion from one fill until the next fill, and cumulative gap days were calculated as the sum of gap days within the 1st year of treatment. Regression analysis covariates: smoking status, COVID period in which measurement took place, insurance type, and average distance to care in patients' geographical area

Baseline Demographics (N=215)



RESULTS

Figure 1. Study Attrition

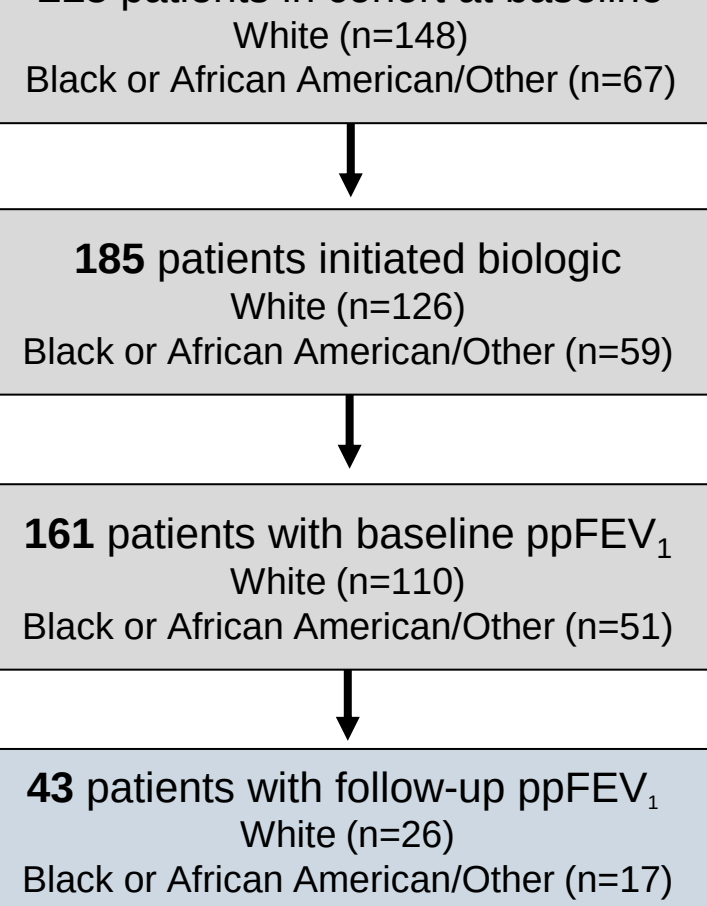
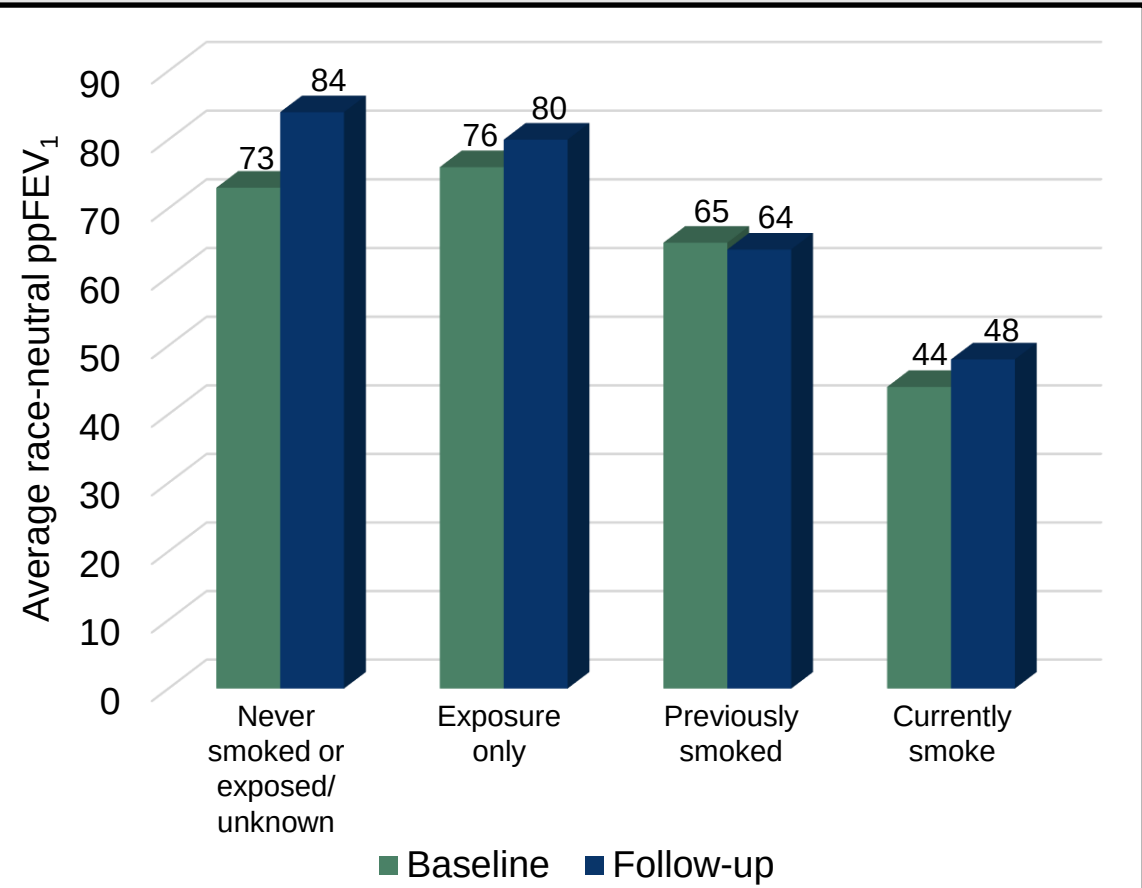


Figure 2. ppFEV₁ by Smoking Status Before and After Biologic Initiation



Smoking Status

Figure 3. Smoking Status by Social Deprivation Index

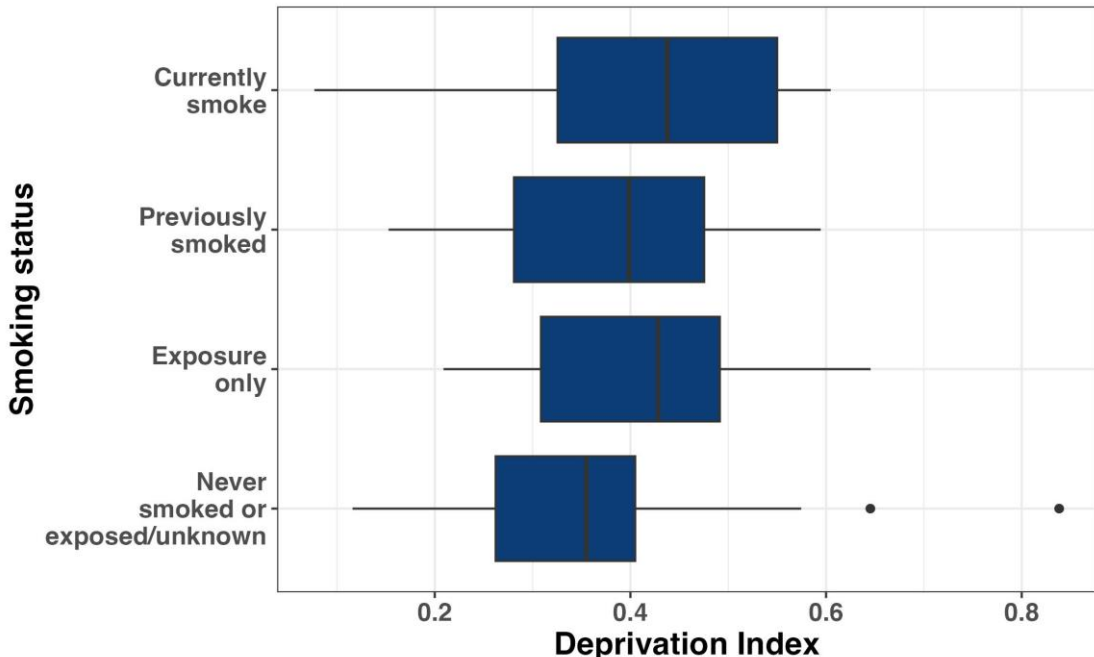
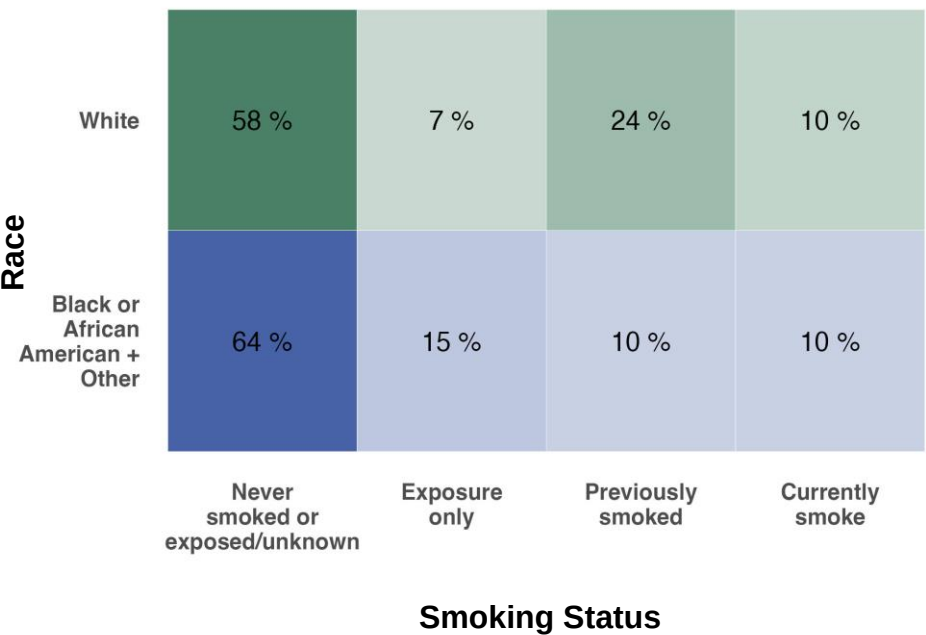
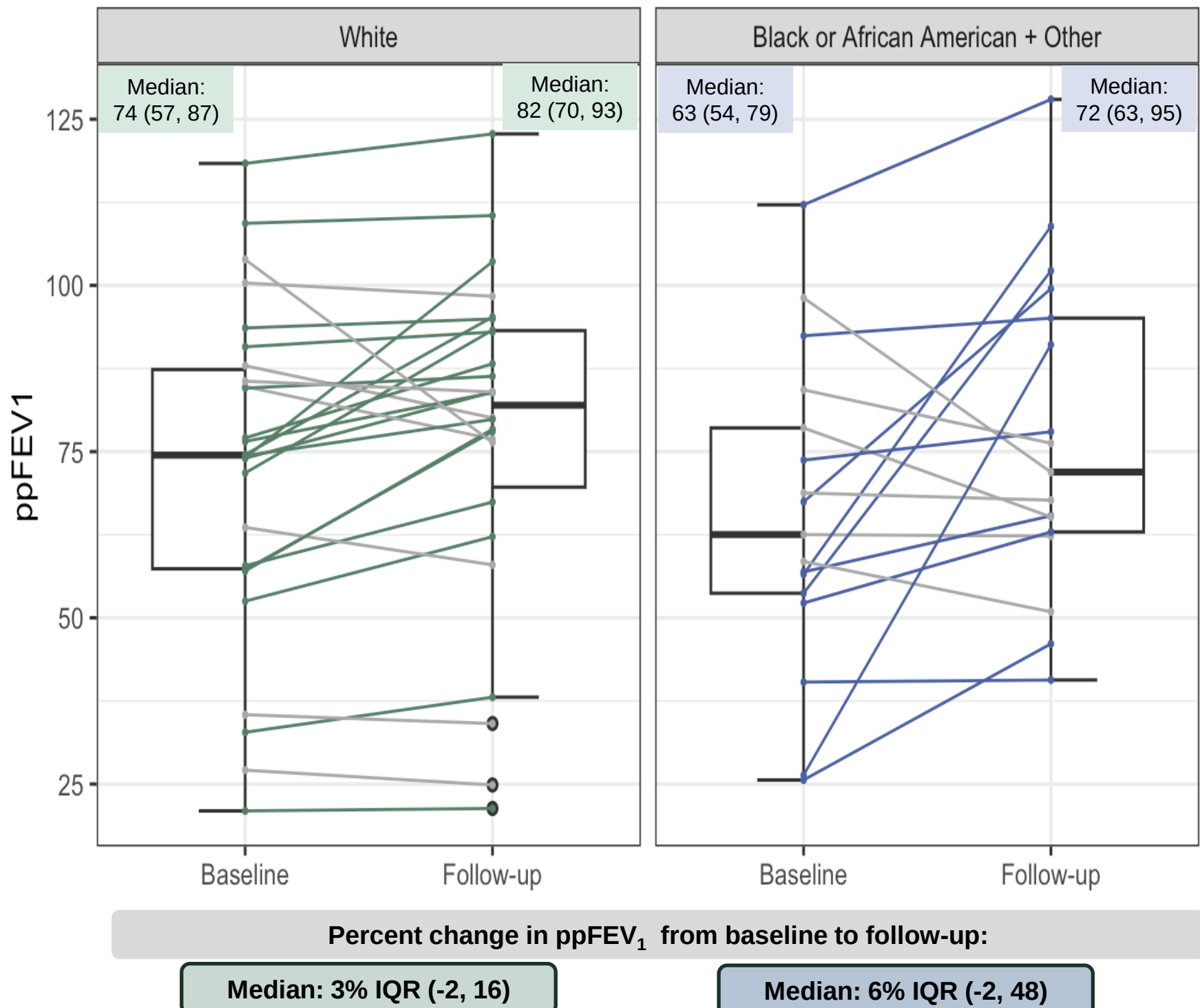


Figure 4. Smoking Status by Race



- Previous (p=0.02) and current smokers (p=0.01) had significantly lower estimated ppFEV₁ at baseline.
- Current smokers in this study were associated with higher social deprivation index scores, indicating worse socioeconomic status.
- 20% of Black or African American/Other patients and 34% of White patients were previous or current smokers.

Figure 5. ppFEV₁ by Race

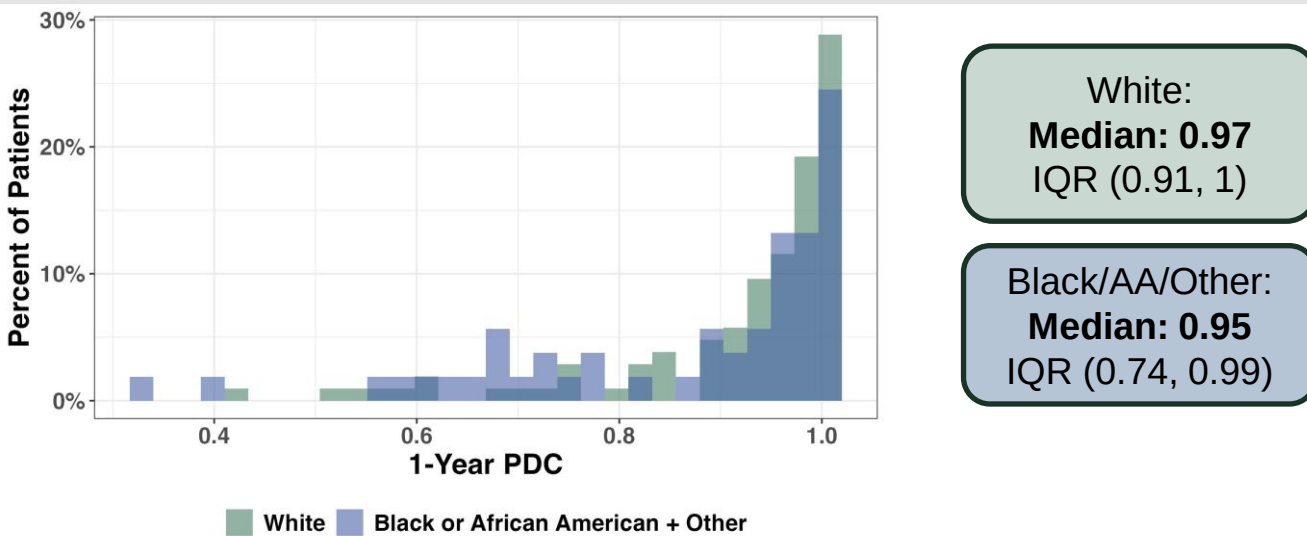


BASELINE:
Black or African American/Other: ppFEV₁ 6.87 less than White patients (p=0.06) at baseline

FOLLOW-UP:
Race was not significantly associated with ppFEV₁ (p=0.71)

Adherence and Gap Days

Figure 6. Adherence (PDC)



Median biologic adherence for all patients was 0.96 (IQR 0.88, 1) within the first year. There were no significant differences in PDC by race.

Figure 7. Gap Days

